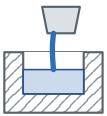
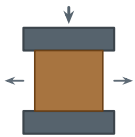
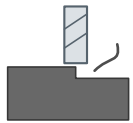
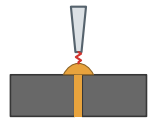
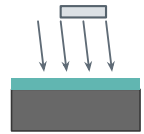
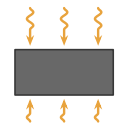
	Shape created	Shape changed				Material changed
	Cohesion created	Cohesion kept	Cohesion reduced	Cohesion increased		Cohesion at atomic scale
	1 Primary shaping *Urformen*	**2 Forming** *Umformen*	**3 Separating** *Trennen*	**4 Joining** *Fügen*	**5 Coating** *Beschichten*	**6 Properties** *Stoffeigenschaft ändern*
conventional	 casting injection moulding pressing + sintering electroforming	 forging rolling deep drawing bending	 turning, milling drilling, grinding laser cutting EDM	 welding brazing adhesive bonding riveting	 painting thermal spraying PVD, CVD electroplating	 heat treatment hardening HIP shot peening
link with AM	**every AM process**: primary shaping, selective, layer by layer, no mould	bonding by plastic deformation: cold spray, friction stir, ultrasonic	support removal, finish machining, hybrid DED + milling	DED and wire-arc AM grew out of welding; fewer joints	cladding and cold spray: coatings grown into parts; repair	post-processing: stress relief, HIP, ageing, peening